MCT
MANJARA CHARITABLE TRUST
RAJIV GANDHI INSTITUTE OF TECHNOLOGY, MUMBAI

CODEASTRA 2.0

Team Details

Team ID : <TEAM ID>

Team Name : <TEAM NAME>

Leader Name : <LEADER NAME>

Domain Chosen : Cybersecurity

Problem Statement : QR Code Safety Scanner – scans QR codes before opening them and warns users if the destination looks malicious

QR Shield

Know what a QR will make you do — before you do it.

Problem Analysis

24 bn
UPI transactions every month
NPCI, Sep 2026

73 crore
UPI QR codes in circulation
Worldline, 2025

2.7×
QR-code fraud cases in 2 years
14,625 → 39,638
Lok Sabha, FY22→FY24

₹805 cr
lost to 10.6 lakh UPI frauds
in just 8 months
MoF, Lok Sabha, Apr–Nov 2025

HOW PEOPLE GET CHEATED

"Scan to receive money"

Fake buyers & 'customer care' send a QR. Scanning a UPI QR always PAYS — victims enter their PIN.

Swapped shop stickers

Fraudster's QR pasted over the shop's: Khajuraho 2025, Aizawl 2024, Goa ₹49.68 lakh (2026).

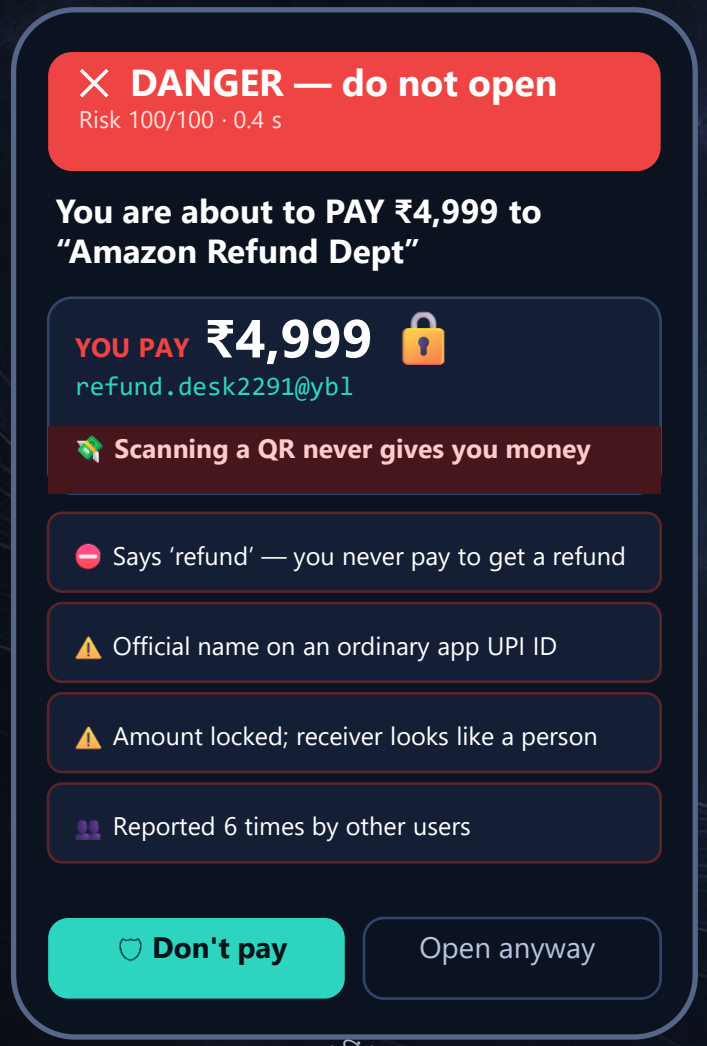
Fake bank / KYC / e-challan

Lookalike sites, AutoPay mandates, APK downloads and *401# call-forward codes.

Root cause NPCI ended P2P 'collect' requests on 1 Oct 2025 — so the QR is now the scammer's main lever. RBI, NPCI and police all say "don't scan unknown QRs", but nobody gives people a tool to check one.

The gap Existing scanners (Kaspersky, Trend Micro, phone cameras) only match links against blacklists. None understand UPI: who you pay, how much, AutoPay — or explain why in Hindi or Marathi.

Proposed Solution & Key Features



UPI-aware decoding

Shows who you PAY, how much, AutoPay mandates; flags lookalike UPI IDs (@okaxisbank, @ybl) & fake 'refund' / 'KYC' notes

"Receiving money?" mode

One tap tells us your intent. Any UPI QR then becomes an instant red stop — kills the #1 scam

Reasons in your language

Plain explanations in English, हिंदी & मराठी — written for shopkeepers & parents, not engineers

Shopkeeper Mode

Register your QR once; a 2-second scan of the counter sticker catches a swap before customers pay

Every QR type

Links, Wi-Fi (fake 'RailWire'), *401# call-forward codes, APKs, AutoPay, Bharat QR checksum

Community + offline

Hashed 'Report this QR' counts; built-in rules still give a verdict with no internet

Technical approach & Innovation

FastAPI server (Python) — the link is never opened on the phone

📱 Phone PWA

Camera / gallery decode
ZXing-C++ WASM
sends only decoded text

→

🧭 Classifier

UPI · Bharat QR · URL
Wi-Fi · tel/SMS · text
AutoPay · app links

→

🔬 Analyzers

UPI semantics & handles
URL lookalikes & intel
Wi-Fi / USSD rules

→

⚖️ Risk engine

Weighted signals → 0–100
SAFE / CAUTION / DANGER
+ reasons in 3 languages

→

✅ Verdict < 3 s

Don't open / Open anyway
Report · History
Shopkeeper check

🧠 Offline rules (always on)

- Unicode TR39 'skeleton' → lookalike letters (onlinesbi = onlinesbi)
- Typo-squats of 50+ Indian banks, govt & brands
- 181 real UPI handles · scam words EN/HI/MR
- .bank.in / .gov.in trust · risky TLDs · APK/USSD

🌐 Online checks (time-boxed ≤ 2.2 s)

- SSRF-safe redirect expansion (IP-pinned, 5 hops)
- RDAP domain age (new = risky)
- OpenPhish · PhishTank · URLhaus feeds
- Google Safe Browsing (optional key)

💡 Innovation / USP

- First scanner that explains UPI before the payment app opens
- Explainable & deterministic — no black box; works offline
- Catches brand-new scams blacklists haven't seen
- Merchant-side sticker verification

TECH STACK

React + TypeScript · Vite PWA · barcode-detector (ZXing WASM) · Python FastAPI · httpx · tldextract + PSL · RapidFuzz · Unicode TR39 · SQLite · pytest (140+ tests)

Impact & Future Scope

TARGET USERS

UPI users & senior citizens
Scan before you pay

Kirana & small shopkeepers
Daily sticker check

OLX/Quikr sellers & students
'Scan to receive' targets

Banks, police & colleges
Awareness drives

IMPACT

- Stops the #1 UPI scam at the moment of scanning — before the PIN
- Non-technical users understand WHY, in their own language
- Shopkeepers catch a swapped sticker before losing a day's sales
- Privacy-first: reports stored as hashes; links never open on the phone

FUTURE SCOPE — ROADMAP

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
SDK / API	**Label OCR**	**1930 & Chakshu**	**More languages**	**On-device mode**
for UPI apps & banks	printed name vs QR payload	one-tap reporting; DoT FRI	+ voice explanations	fully offline rules

Supporting Information

USER WORKFLOW

1 Scan / upload → 2 Decode on phone → 3 Analyze on server → 4 Verdict + reasons → 5 Don't open · Report

✅ Built & validated (working prototype)

- 140+ automated tests: UPI parser, homoglyphs, scoring, SSRF guard
- 14 printable demo QRs — every one gives the expected verdict
- SSRF guard tested vs private IPs, cloud metadata, DNS rebinding, redirect loops
- Verdict in ~0.1 s offline, < 3 s with live checks
- Demo mode: reserved .test domains, works with no internet
- Installable PWA: Android & iPhone browsers

References

- NPCI — UPI Linking Specification v1.6 (upi://pay parameters, signed QR)
- RBI consumer advisory, 28 Jan 2022 — "receipt of money does not require scanning QR"
- Lok Sabha USQ 2245, 9 Dec 2024 — QR-code fraud FY22–FY25
- MoF Lok Sabha reply, Dec 2025 — UPI fraud ₹805 cr (Apr–Nov 2025)
- NPCI circular, 29 Jul 2025 — P2P collect discontinued from 1 Oct 2025
- RBI circular, Apr 2025 — exclusive .bank.in domain for banks
- DoT advisory, Jan 2024 — *401# call-forwarding scam
- Unicode TR39 confusables · OWASP SSRF Prevention Cheat Sheet
- Interisle Phishing Landscape 2025 · Unit 42 newly-registered-domain study

LIVE DEMO SCENARIOS — every QR on our printed sheet gives the expected verdict

SBI .bank.in | Kirana UPI | Home Wi-Fi | Café open Wi-Fi | 'Refund' UPI | @okaxisbank KYC | AutoPay trap | onlinesbl.test

onlinesbi (Cyrillic) | Short link → fake HDFC | e-challan APK | Fake RailWire | *401# code | Swapped sticker